

Plant 2002 QC Master Data

Every step needed to take SAP QM in plant 2002 from empty to a working inspection lot, in the order the system enforces - with the traps that cost time the first time through.

Field	Value
System	KSD, client 500
Plant	2002 Kejriwal Ind. Pvt. Ltd. - Dyeing
Company code	2000
Workbench transport	KSDK906754 (task KSDK906755) - CDS views, behaviour, apps
Customizing transport	KSDK906757 (task KSDK906758) - the UD selected set
Written	29 August 2026, from work carried out in KSD

Master data does not transport

Only the usage-decision selected set travels on the Customizing request. Master inspection characteristics, the sampling procedure, inspection plans and material QM views are **master data** - they exist per system and must be built again in KSQ and in production. This runbook is therefore written to be followed more than once.

The order is enforced, not suggested

Each step depends on the one above it, and SAP refuses the later ones if an earlier one is missing. The refusals are not always obvious - step 5 in particular fails with three different errors in sequence.

- 1 Usage decision selected set ZQC-UD Customizing
- 2 Master inspection characteristics QS21 master data
- 3 Sampling procedure ZQC-FIX master data
- 4 Inspection plans QP01 needs 2 and 3
- 5 Material QM view MM01 needs 4 for a full setup
- 6 First inspection lot QA01 or a goods receipt

Step 1 - Usage decision selected set

The nine outcomes a technician can pick at the usage decision, and - the part that matters - whether each one means accept or reject.

Route

Catalog type 3 is Customizing, so **QS51 refuses it**. Use either:

SPRO -> Quality Management -> Quality Inspection ->
Inspection Lot Completion -> Edit Selected Sets

or SM30 -> V_QPAM_UD (the set header)
SM30 -> V_QPAC_UD (the nine codes)

Assign to the **Customizing** request, not the workbench one.

Header

Field	Value
Plant	2002
Catalog	3
Selected set	ZQC-UD
Status	2 Released
Short text	QC usage decision - yarn
Def. Group / Def. Code	leave blank

Status is off-screen. In SM30 the Status column sits to the right of Short Text and you must scroll the table sideways to reach it. Leave it unset and the row stays red with *Specify a status*, which does not name the column it means.

Default group and default code stay blank on purpose. A default usage-decision code pre-selects an outcome for the technician, which is the opposite of what a quality gate is for.

The nine codes

Code group ZQC-UD on every row.

Code	Valuation	Score	Meaning
A	A accept	100	Accept
A1	A accept	90	Accept with deviation
SP	A accept	80	Special acceptance
ST	R reject	70	Reject
DG	R reject	60	Reject
PQ	R reject	60	Reject
CLQ	R reject	50	Reject
RD	R reject	50	Reject
JL	R reject	40	Reject

The valuation column is the one that carries weight. QAVE-VBEWERTUNG is what the usage decision writes to the lot, and it is what the apps read to decide whether a lot passed. The code itself is never hardcoded

anywhere. Get A, A1 and SP marked **A** and the other six **R**, and the gate behaves correctly whichever code a technician picks.

Underscores are not allowed in catalog keys. Hyphens are - which is why the name is ZQC-UD and not ZQC_UD. This matters again in step 4.

Step 2 - Master inspection characteristics

Nineteen characteristics in QS21, plant 2002. These are measurement *definitions* - what is measured, in what unit, to how many decimals. They deliberately carry **no limits**.

Do not bake limits into the characteristic. Denier tolerance for a 150/48 yarn is not the tolerance for a 300/96. Limits belong on the inspection plan, per material. Put them here and you need nineteen characteristics per yarn count.

Working order - and it matters

- 1 QS21 -> name, plant 2002
- 2 Short Text, Search Field
- 3 Control indicators, both pages
- 4 Decimal places and unit
- 5 Status -> Released <-- LAST
- 6 Save

Release last. A released characteristic cannot be edited in place - SAP spawns a new version instead. Setting the status before the decimals leaves you with two same-dated versions of the same characteristic and no clean way to tell which one a plan will pick up.

General data

Field	Value
Plant	2002
Short Text	the characteristic in plain words
Search Field	a searchable word - DENSITY, PACKAGE, TENACITY
Status	Released (set last)
Copy model	Incomplete Copy Model - leave as defaulted
Preset Indicators	leave blank
Quantitative Charc	ticked

Incomplete Copy Model is what allows the characteristic to be saved with no target and no limits. Switch it to a complete model and SAP demands them here, which is exactly what we are avoiding.

Preset Indicators applies a bundle of control indicators from Customizing and **overwrites** anything set by hand. If your system forces a value, re-open the control indicators afterwards and check all five survived.

Control indicators - identical on all nineteen

Page 1
Lower Specif. Limit ticked
Upper Specif. Limit ticked
Check Target Value ticked
Sample all clear
Summ. Recording selected
Required Charc selected

Page 2
Fixed Scope selected
No Documentation selected
Record Measured Vals ticked
Print selected
No Formula selected

Setting	Why
Summ. Recording	The apps read QAMR. Single-value recording writes elsewhere and the results screens come up empty.
Record Measured Vals	Writes the actual numbers. Without it you get a pass or fail with no measurement behind it.
Required Charc	A technician cannot confirm results while leaving the characteristic blank.
Sample - all clear	Ticking Sampling Procedure here makes one mandatory on the characteristic. Leave it clear and the plan assigns it.
Fixed Scope	Sample size comes from the plan. The inspector cannot quietly test three cheeses instead of ten.

Units - what this system actually has

Checked against T006A. The internal key and the displayed code differ for one of them.

Displayed	Internal key	Used by
%	%	ELONG, BWS, OPU, MOISTURE, EVENNESS, CRIMP
mm	MM	TRAVERSE, PKG_DIA
kg	KG	PKG_WT
g/cm3	RHO	PKG_DENS
g	G	WND_TENS
DEN	DEN	DENIER, DEN_NOM

cN/dtex does not exist, and neither does any tex- or centinewton-based unit. TENACITY therefore carries **no unit**, and cN/dtex goes in its Short Text instead so the technician still sees it. Creating the unit properly in CUNI is possible but MSEHI is only three characters, so it would be a key like CDT displaying cN/dtx, and it needs a dimension decision. Worth doing only if the unit is wanted on QC certificates.

The nineteen characteristics

MIC	Measures	Unit	Dec	Standard
DENIER	Measured linear density	DEN	1	ISO 2060 / ASTM D1907
DEN_NOM	Nominal denier from the material	DEN	0	from material master
ELONG	Elongation at break	%	1	ISO 2062 / ASTM D2256
TENACITY	Tenacity (cN/dtex)	-	2	ISO 2062 / ASTM D2256
FILAMENT	Filament count	-	0	ASTM D2259
BWS	Boiling water shrinkage	%	2	ISO 18067
OPU	Oil pick-up	%	2	ISO 18066
MOISTURE	Moisture regain	%	2	ISO 6741-1
EVENNESS	Evenness / Uster CV%	%	2	ASTM D1425
NIPS	Interlace nips per metre	-	0	ASTM D4724
CRIMP	Crimp contraction	%	1	ISO 16549
PKG_DENS	Package density	g/cm3	3	-
PKG_HARD	Package hardness	-	0	-
TRAVERSE	Traverse / package length	mm	0	-
PKG_DIA	Package diameter	mm	0	-
PKG_WT	Package net weight	kg	3	ISO 5688
DELTA_E	Shade difference, delta E CMC(2:1)	-	2	ISO 105-J03
BRK_FIL	Broken filaments per package	-	0	visual count
WND_TENS	Winding tension	g	1	-

The three in bold were found later, by reading the deployed app screens against the master data: each is a field a technician can type into with nothing behind it to save the reading. They are as necessary as the other sixteen.

Step 3 - Sampling procedure

One procedure, QDV1, name ZQC - FIX. It is client-level, so plant 2002 picks it up without a plant field.

Field	Value
Sampling Type	Fixed sample
Valuation Mode	Attributive inspection nonconf. units
Control Chart Type	blank
Determination Rule	10 Fixed sample
Inspection Points	Without
Multiple Samples	No multiple samples

Why that valuation mode

It counts how many sampled packages fall outside the plan's limits and fails the characteristic when the count exceeds the acceptance number. Set acceptance to 0 and any single out-of-spec reading fails it.

Rejected alternative	Why not
Mean value within tolerance range	Valuates the average. Five good cheeses and one badly off can average inside the limits and pass - the exact defect the gate exists to catch.
Variable insp. s-Method	Statistical lot acceptance from mean and standard deviation. It can reject a lot where every individual reading is in spec, and accept one where some are not. Correct for its purpose, baffling for a technician who can see the numbers.
The two / <i>manual</i> variants	Valuate automatically but let the inspector overrule. That undercuts the gate. The human judgement call belongs at the usage decision, where ZQC-UD records why.

Leave Control Chart Type blank. Every entry in that dropdown is an SPC control chart. Choosing one turns this into statistical process control, which expects a run of historical data to compute control limits from, and characteristics start being flagged on trend rather than against your spec limits.

With determination rule 10, the **sample size and acceptance number live on the plan characteristic**, not here. One procedure therefore serves all nineteen characteristics, and you can test ten cheeses for denier but two for package weight.

Step 4 - Inspection plans

Three plans in QP01, one per gate. This is where the specification finally lands, and it is the only step that cannot be done from system knowledge alone.

Plan	Inspection type	Raised when	Characteristics
Greige	01	Goods receipt of greige yarn	DENIER, DEN_NOM, ELONG, TENACITY, FILAMENT, BWS, OPU, MOISTURE, EVENNESS, NIPS, CRIMP
Post-Dyeing	03	Production order release	DENIER, ELONG, TENACITY, BWS, MOISTURE, DELTA_E
Post-Winding	04	Production order release	DENIER, MOISTURE, BRK_FIL, WND_TENS, PKG_DENS, PKG_HARD, TRAVERSE, PKG_DIA, PKG_WT

Header: usage 5, status 4 released. Assign materials inside QP01.

Per characteristic

Master inspection characteristic from the list in step 2
Sampling procedure ZQC-FIX
Target value from the specification
Lower limit / Upper limit from the specification
Sample size how many packages the lab tests
Acceptance number 0 = any out-of-spec reading fails

The numbers are yours and cannot be invented. A guessed denier tolerance either passes bad yarn or rejects good yarn, and either way the gate is worse than no gate. Collect them on the specification workbook that accompanies this document before starting this step.

Still to decide at this step

Open item	What is needed
Shade outcomes	The post-dyeing app reads a list of allowed shade outcomes from a catalog type 1 selected set. Decide the terms your dyeing master uses, then create the set. Note the app currently filters on ZQC_GREY with an underscore, which SAP will not accept as a catalog key - it must become ZQC-GREY, and the app changed to match.
Defectives count	All three apps show a <i>Defectives found ... of sample</i> control. That is an attributive count, not a quantitative measurement, and nothing in the plan records it. Decide whether it becomes a characteristic or goes through a defect catalog.

Step 5 - Material QM view

Activating the inspection type on each material. This is the step with the most traps, and none of them announce themselves clearly.

MM02 will not show you the Quality Management view if the material does not already have one. MM02 offers only the views a material already has - so on a material with no QM data, the view is simply absent from Select View(s), along with MRP and Work Scheduling if those are missing too. This looks exactly like a material type configuration problem and is not one.

Use **MM01** instead. It recognises that the material exists and lets you extend it with a new view.

```
/nMM01
Material the existing material number
Industry must match the existing material
Material Type must match the existing material
Copy from LEAVE BLANK

Select View(s) tick ONLY Quality Management
Org. Levels plant 2002
Insp. Setup new entry
```

Tick one box only. MRP 1-4, Forecasting and Work Scheduling sit in the same list. Ticking one creates an empty view on a material that is live in production orders, and empty MRP data changes how it is planned.

Leave Copy from blank. It sits right under the fields you are filling and is tempting. Copying from another material pulls that material's data into yours.

QA08 is not an alternative for the first setup - it finds materials that already have QM data and cannot create the first record. It is useful afterwards, for changing many materials at once.

Inspection setup flags

SAP validates these as a dependency chain and refuses them one at a time, in order. Each error names a different thing and none of them says *you have no inspection plan*.

Flag	With a plan	Before a plan	If wrong
Active	ticked	ticked	
Insp. with Task List	ticked	clear	<i>You can only inspect by characteristic using the inspection specification</i>
Inspect Charcs	ticked	clear	same error, one link further down
Automatic Assignment	ticked	clear	<i>Automatic assignment of specs not possible</i>
Post to Insp. Stock	your choice	clear	-
Automatic UD	clear	clear	-

Leave Automatic UD clear, permanently. It makes the material eligible for the automatic usage-decision job QA40 - a background job that closes inspection lots without anyone looking at them. On a gate whose whole purpose is that a technician decides, and where passing posts a real goods movement, this is the one setting that would quietly bypass everything.

Post to Insp. Stock routes goods receipts into quality-inspection stock, so a rejected lot physically cannot be consumed. That is the strongest form of the gate and the right end state for greige - but turn it on deliberately, because it changes behaviour for everyone receiving that material.

A warning you can accept

Check the data for sample calculation for inspection type NN is a warning, not an error. With no characteristics there is nothing to sample. Press the tick and carry on, or put ZQC-FIX in the Smpl. Procedure field to silence it.

Step 6 - First inspection lot

In normal running the lot appears by itself - a goods receipt raises type 01, a production order release raises 03 and 04. To test before that, create one by hand.

```
/nQA01
Material an activated material
Plant 2002
Insp. Lot Origin 01 Goods Receipt

Batch a REAL batch - check MCHA first
Insp. Lot Qty 100 Unit KG
Supplier a vendor who actually supplied it
Purchasing Org. 2002
```

No purchase order, no goods movement, no stock touched. Origin 01 and inspection type 01 both being 01 is correct, not a duplication - one is where the lot came from, the other is what kind of inspection it is.

Use a batch that exists. Manual lot creation does not validate the batch hard, so a typo either sits against a batch that is not there or quietly creates one. Check MCHA for the material first - and watch for look-alikes: AT1-5547 and AT01-5547 are not the same string.

Sample size will come out 0.000 if the material has no task list yet. That is expected - no characteristics, nothing to sample.

What to check afterwards

Check	Expected
Lot appears in the app worklist	One row, with the batch in the Greige lot column
Production lot column	<i>not assigned yet</i> unless a plant batch record claims the batch
Production order	Filled if the material has exactly one order in the last 365 days - the app resolves it by elimination and says so
Characteristics list	Empty until the inspection plan exists
Browser shows old columns	Clear the cache - see the note below

After any app deploy, clear the browser cache before judging the result. UI5 bundles every view into Component-preload.js, so a normal reload keeps rendering the old build even after a successful upload. DevTools -> Network -> Disable cache, then reload with it open. Fetching a view XML directly from the server distinguishes a stale client from a failed upload.

Traps, collected

Every one of these cost time the first time through.

Symptom	Cause and fix
QS51 refuses the selected set	Catalog type 3 is Customizing. Use SPRO or SM30 on V_QPAM_UD / V_QPAC_UD.
Row stays red, <i>Specify a status</i>	The Status column is off-screen right in SM30. Scroll the table sideways.
Two versions of the same characteristic	It was Released before the decimals were set. A released MIC cannot be edited in place. Set everything first, release last.
Control indicators reverted	Preset Indicators was filled and overwrote them. Leave it blank, or re-check the five settings after.
CN/DTEX not in the unit list	It does not exist in this system. Leave TENACITY unitless and put the unit in its Short Text.
g/cm3 not found by that name	Its internal key is RHO. Pick it from F4 by the g/cm3 display.
Quality Management missing from MM02	MM02 shows only views the material already has. Use MM01 to extend.
QA08 says <i>no entry found</i>	QA08 finds existing QM records and cannot create the first one. Use MM01.
<i>You can only inspect by characteristic...</i>	Inspect Charcs is ticked with no task list. Untick it.
<i>Automatic assignment of specs not possible</i>	Automatic Assignment is ticked with no specs to assign. Untick it.
App worklist shows old columns	Stale Component-preload.js in the browser. Disable cache and reload.
Filtering by Supplier Lot finds nothing	Supplier Lot is QALS-LICHN, the vendor's own reference - not the batch. A manually created lot has none.

One finding that is not a trap

ZPP_BATCHN - LOTNO is **not a lot number**. It was one until mid-2013 and the plant then stopped filling it in; since then it holds the yarn quality - BRT alone covers 15,096 rows spanning 2002 to 2026. Plant-wide there are 410 distinct values over 131,562 rows.

Nothing in this system records which supplier lot fed which plant batch. MSEG-AUFNR is blank on every 301 transfer, and the 301's receiving batch is a yarn-quality bucket rather than a lot - DRM1 holds 745 real supplier lots, DPR1 holds 33 quality codes. So the greige inspection lot sits on the **supplier batch in DRM1**, which is the only identifier in the system that describes real yarn, and the production order is resolved by elimination rather than from a recorded link.

This is why the apps ask the operator to pick an order rather than resolving it silently, and why the Production lot column says *not assigned yet* instead of echoing something back.

Where plant 2002 stands

Step	State
1 ZQC-UD selected set	Done - nine codes, valuations verified in QPAC
2 Master inspection characteristics	Sixteen done. DELTA_E, BRK_FIL, WND_TENS still to create
3 ZQC-FIX sampling procedure	Done
4 Inspection plans	Not started - waiting on the specification workbook

Step	State
5 Material QM view	One material activated for type 01, minimal settings
6 First inspection lot	Lot 010000000051 created and rendering in Grey QC
Lot 010000000051 resolved its production order to 000001008691 by elimination, showed the greige lot AT1-5547 correctly and left the production lot blank - the intended behaviour on all three counts. What it cannot yet do is record a measurement, because the inspection plans do not exist.	